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Paleomagnetic evidence for oroclinal bending of the southern Antarctic Peninsula

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ABSTRACT

Paleomagnetic results from the investigation of 13 magnetically stable units (92 oriented rock samples) of Upper Cretaceous ("Andean") plutons and dikes from the Orville Coast and eastern Ellsworth Land, Antarctica, define a mean direction of magnetization of $I = -77°$, $D = 51°$ ($\alpha_{95} = 5.9°$), with a paleomagnetic pole at 71°S, 165°W. The sampled units were emplaced after the Late Jurassic to Early Cretaceous intense folding associated with subduction along the western side of the Antarctic Peninsula. In addition, all sampled intrusive rocks are normally magnetized and are believed to have been emplaced during the Late Cretaceous period of predominantly normal polarity. There is no evidence of post-emplacement remagnetization. Unlike rocks from other Andean paleomagnetic collecting localities on the Antarctic Peninsula, whose mean declinations are oriented approximately north, the mean declination of samples from the Orville Coast and eastern Ellsworth Land is rotated 51° clockwise from north. Uncertainty in declination at the 95% confidence level (δ_{95}) is ±27°. The data support the conclusion that the southern bend of the S-shaped Antarctic Peninsula was formed after Late Cretaceous time. Early Tertiary right-lateral transform faulting across the base of the Antarctic Peninsula may have produced this major oroclinal bend.

Data from four localities (21 samples; $I = -79°$, $D = 30°$, $\alpha_{95} = 4.3°$) in Upper Jurassic(?) massive rhyodacite porphyry lava flows in the northern part of the area are similar to those of the Andean igneous rocks. Although the evidence is not conclusive, it seems most probable that the porphyry was magnetically reset by Late Cretaceous plutonism.

INTRODUCTION

Paleomagnetism has been employed in many studies as a tool for measuring large-scale bending or rotation of tectonically active regions of Earth's crust. Investigations such as those of Van der Voo (1969) on the counterclockwise bending of the Iberian Peninsula and Kawai and others (1969) on the oroclinal bending of Japan have added greatly to our understanding of plate-tectonics processes. Other studies also have been made to measure oroclinal bending of the Antarctic Peninsula, an S-shaped arc of predominantly Mesozoic sedimentary and volcanic rocks intruded by upper Mesozoic to Cenozoic plutons of the "Andean intrusive series" (Adie, 1955; Elliot, 1975). Hamilton (1966) contended that the paleomagnetic data of Blundell (1962) indicate significant post-Cretaceous structural distortion of the northern part of the peninsula. However, on the basis of additional data, Kellogg and Reynolds (1978) supported the conclusions of Dalziel and others (1973) that no such post-Cretaceous structural bending of the peninsula could be detected from the existing paleomagnetic evidence. This result has been supported more recently by Valencio and others (1979). Kellogg and Reynolds calculated a 95% confidence limit on mean declination (δ_{95}) for a group of paleomagnetic directions from the expression $\delta_{95} = \sin^{-1}(\sin \alpha_{95}/\cos I)$, where α_{95} is the semiangle of the cone of 95% confidence and I is the mean inclination. Such calculations, especially at high latitudes, are imperative for determining uncertainty in declination and therefore for determining any possible oroclinal bending. One important conclusion from the study by Kellogg and Reynolds is that at paleolatitudes higher than about lat 65° to 70°S, large inclinations in paleomagnetic direction result in uncertainties in declination that are too large for meaningful determination of oroclinal bending. Consequently, it was anticipated that the paleomagnetic results from the Orville Coast and eastern Ellsworth Land (Fig. 1), the region sampled during the study described here, would not be particularly useful for discerning such structural bending. Surprisingly, this was not the case; enough oriented samples from a variety of localities were collected to unambiguously measure significant oroclinal bending.

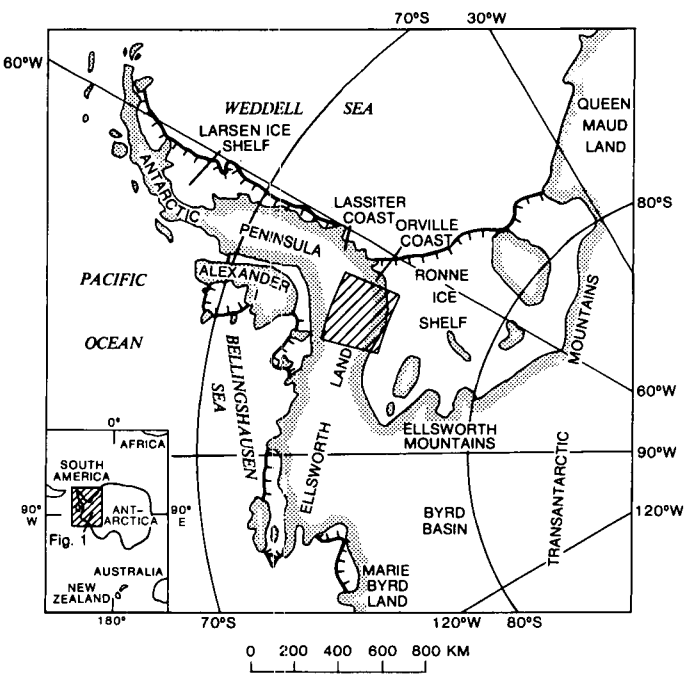


Figure 1. Location of Orville Coast (diagonal-ruled area) and part of eastern Ellsworth Land, Antarctica. All sampling sites are within ruled area. Hachured line shows extent of shelf ice.

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GEOLOGY OF THE ORVILLE COAST AND EASTERN ELLSWORTH LAND

The largely ice-covered region studied lies at the extreme southern end of the Andean cordillera (Fig. 1) and consists of highly folded Middle to Upper Jurassic mudstone, siltstone, and sandstone of the Latady Formation, and silicic to intermediate calc-alkalic volcanic rocks that are in part interbedded with the Latady Formation in the northern part of the region (Laudon and others, 1964, 1969; Laudon, 1972; Rowley, 1978; Thomson and others, 1978). Suarez (1976) interpreted the Latady Formation to have been deposited in a Middle to Upper Jurassic backarc environment on the Weddell Sea side of the main volcanic arc. A massive porphyry unit, rhyodacitic to dacitic in composition and at least 1,000 m thick in places, was first described by Laudon and others (1964) and occurs extensively in the northern regions of the Orville Coast and eastern Ellsworth Land. This unstratified rock, apparently part of the Mesozoic magmatic arc complex, probably represents thick lava flows, ash-flow tuffs, or flow domes. The age of the massive porphyry is open to some debate; it is almost certainly older than Upper Cretaceous plutonic rocks in the area, and it represents at least part of the source rock for the Latady Formation sediments (Thomson and others, 1978).

The sedimentary rocks and layered volcanic rocks are strongly folded about axes that parallel the shape of the peninsula (Rowley, 1978). The strikes of fold axes swing from about N60°E in the northeast, adjacent to the Lassiter Coast, to about N70°W in eastern Ellsworth Land. The period of folding occurred sometime between Late Jurassic and Late Cretaceous time, in response to forces generated at an active subduction zone located to the northwest (for example, Herron and Tucholke, 1976). The Latady Formation and the volcanic rocks were intruded, partially refolded, and contact metamorphosed by Upper Cretaceous plutons of gabbroic to granitic composition (Laudon and others, 1969; Laudon, 1972; Rowley and Williams, 1974; Rowley, 1978). Published K-Ar and Rb-Sr biotite and hornblende ages from 13 plutons and two dikes from the Lassiter Coast region range from 95 to 120 m.y. (Farrar and others, 1980), in excellent agreement with the 102- to 109-m.y. ages obtained from Rb-Sr whole-rock ages from five plutons in eastern Ellsworth Land (Halpern, 1967). Two of the dated plutons in eastern Ellsworth Land (locs. Ke189 and Ke190 of this study) were sampled for paleomagnetic study and yielded acceptable results.

Numerous felsic to mafic dikes intrude all the units mentioned and are interpreted to be an end phase of Late Cretaceous plutonism. Some of these dikes are nearly identical petrographically to dated Upper Cretaceous dikes from the Lassiter Coast (P. D. Rowley, 1978, oral commun.). Four dikes were sampled for paleomagnetic measurement and gave acceptable results. Two of these dikes (Ke127 and C15) intrude the Latady Formation, one (Ke141) cuts a gabbro pluton, and one (Ke145) intrudes the massive porphyry.

PALEOMAGNETIC METHOD

One hundred and twenty-seven hand samples, oriented by magnetic compass and hand level, were collected from 19 different sites: nine from separate plutons, four from dikes, five from the massive porphyry, and one from a steeply dipping lava flow. Mafic inclusions were sampled from three of the more felsic coarse-grained plutons to avoid the problem of collecting samples with large, possibly magnetically unstable multidomained grains of titanomagnetite. The variety of rock types collected clearly represents a span of time far longer than is needed to average secular variation.

With two exceptions (locs. Ke101 and V132), at least six oriented samples were collected at each locality. Corrections in magnetic declination were periodically checked in the field by foresight-backsight methods against the sketch map of the area compiled by the U.S. Geological Survey (1969). One oriented core was extracted from each sample in the laboratory, and directions of magnetization were measured on a 10-Hz spinner magnetometer. After measurement of the natural remanent magnetization (NRM), samples were cleaned in peak alternating fields (af) of 50 oe, 100 oe, 300 oe, and 400 oe. Several sets of samples were cleaned in fields as high as 600 oe. In addition, to check for any possible thermal overprinting, 22 arbitrarily selected samples from eight localities, representing all major rock types, were thermally demagnetized in steps to temperatures of 550°C and compared with the results of the af demagnetization.

The af cleaning level that produced the smallest α_{95} was chosen to represent a locality. Consistent changes in mean direction during cleaning, indicating the differential removal of a low-coercivity component, were not observed. With only two exceptions, Ke109 in the massive rhyodacite porphyry and Ke188 in the layered pyroclastic rocks, the rocks from all sample localities defined an α_{95} of less than 15° and were therefore defined to be paleomagnetically acceptable. A Fisher statistical analysis was applied to the data (Fisher, 1953), and a virtual geomagnetic pole for each paleomagnetically acceptable locality was calculated.

The overall mean directions and their limits on uncertainty (α_{95}) were calculated by the method outlined by Watson and Irving (1957). By this analysis, the overall mean direction from one rock group (for example, all dikes) was estimated from all sample directions from acceptable sites in that group. The analysis of dispersion used to determine the overall values for α_{95} incorporates a weighted average of the number of samples from each site, as well as taking into account both the precision calculated at each collecting locality and the precision calculated about the overall mean direction.

The analysis of dispersion employed by Kellogg and Reynolds (1978) in an investigation of various paleomagnetic studies from the Antarctic Peninsula (Blundell, 1962; Dalziel and others, 1973; Scharnberger, 1980) did not incorporate the method of Watson and Irving (1957). Dispersion about the overall mean direction from each study was calculated essentially by giving equal weight to each sample direction. Consequently, the value for α_{95} (and therefore δ_{95}) thus obtained was too small. This fact, however, in no way invalidates the conclusions of the Kellogg and Reynolds paper, but simply gives more leeway for possible oroclinal bending along the peninsula.

PALEOMAGNETIC RESULTS

Of the 19 localities sampled, data from 17 proved to be paleomagnetically acceptable (Table 1). These are from four dikes and nine plutons of various composition and from four localities in the massive porphyry. Figure 2 shows both the NRM directions and the directions at the best demagnatization level for all samples from each of these rock groups.

All acceptable localities on the Orville Coast and eastern Ellsworth Land displayed normal polarity. Similar findings have been reported by Kellogg and Reynolds (1978) and by Scharn-

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berger (1980) from other Andean intrusive rocks farther north on the peninsula. This uniform normal polarity strongly suggests that plutons, dikes, and conceivably the porphyry unit were emplaced during Late Cretaceous time when the geomagnetic field was predominantly of normal polarity (Keating and others, 1975).

It might be argued that all Andean rocks were reset during some large thermal event that occurred during a period of normal polarity. The following five pieces of evidence, however, collectively contradict this possible history:

1. No consistent change in mean direction, indicating the differential removal of a secondary lower coercivity component, was observed during af demagnetization of any unit.

2. During both af and thermal demagnetization, magnetic intensities showed a decrease in magnitude. An increase in intensity

TABLE 1. SUMMARY OF ACCEPTABLE PALEOMAGNETIC DATA, ORVILLE COAST AND EASTERN ELLSWORTH LAND, ANTARCTICA

Site number and rock description	Location		N	Demagnetization field (oe)	I	D	R	k	α_{95}	Paleomagnetic pole		dm	dp
	lat (°S)	long (°W)								lat	long		
Ke101, coarse-grained granodiorite	75.3	70.2	5	50	−66	51	4.988	332	4	56°S	177°E	7	6
Ke127, dacite dike in Latady Formation	76.0	69.6	6	400	−80	56	5.946	92	7	74°S	150°W	13	13
Ke141, fine-grained mafic dike in gabbro (Ke142)	75.2	69.5	8	200	−68	51	7.784	33	10	58°S	179°E	17	14
Ke142, coarse-grained gabbro	75.2	69.5	8	200	−65	44	7.930	100	6	57°S	170°E	9	7
Ke145, fine-grained mafic dike in porphyry	75.2	69.2	6	400	−81	22	5.902	51	6	83°S	176°W	18	18
Ke154, mafic inclusions in granodiorite	75.2	68.8	6	50	−69	67	5.844	32	13	56°S	162°W	22	19
Ke157, medium-grained gabbro	75.1	68.1	6	200	−68	94	5.970	169	5	48°S	137°W	9	7
Ke158, mafic inclusions in granodiorite	75.3	68.1	7	150	−74	24	6.725	22	13	72°S	155°E	24	21
Ke189, coarse-grained diorite*	75.1	72.0	12	50	−88	221	11.846	71	5	72°S	63°W	10	10
Ke190, mafic inclusions in granodiorite†	75.4	72.8	8	50	−74	335	7.927	96	6	73°S	63°E	10	9
C15, fine-grained andesite dike in Latady Formation	76.0	69.7	8	300	−85	94	7.796	34	10	72°S	105°W	19	19
Ro410, medium-grained diorite	75.9	70.4	6	200	−77	78	5.860	36	11	65°S	141°W	21	20
L31, medium-grained diorite	75.2	69.6	6	300	−64	61	5.826	29	13	51°S	172°W	20	18
Massive rhyodacite porphyry													
Ke147	75.2	69.3	6	200	−77	30	5.862	36	12	76°S	170°E	21	20
Ro445	75.2	69.1	6	200	−81	25	5.965	143	6	82°S	174°W	11	10
By51	74.3	73.1	6	100	−76	−26	5.972	176	5	75°S	159°W	9	9
V132	75.1	69.4	3	200	−88	37	2.995	425	6	79°S	83°W	12	12
All plutons (9 sites)	75.3**	70.0**	64	..	76.9	49.3	61.233	22.8	5.9††	71.1°S	150.9°W	11.0	10.2
All dikes (4 sites)	75.6**	69.5**	28	..	76.9	56.1	27.147	31.6	7.2††	69.6°S	160.7°W	13.4	12.5
All massive porphyry units (4 sites)	75.0**	70.2**	21	..	−79.1	29.9	20.739	76.5	4.3††	79.0°S	179.8°W	8.2	7.8
All plutons plus dikes	75.4**	69.8**	92	..	−76.9	51.4	88.398	25.3	5.9††	70.8°S	164.6°W	11.0	10.2

Note: Data from two sites, a massive rhyodacite porphyry (Ke109) and a steeply dipping lava flow (Ke188), proved too scattered to be considered acceptable. Refer to text for the meaning of "acceptable."

N is the number of specimens measured; demagnetization field is that at which the smallest α_{95} occurred; I is mean inclination, negative upward; D is mean declination, clockwise from true north; R is the magnitude of the vector sum of the N unit sectors; k is Fisher's (1953) best estimate of precision; α_{95} is the semiangle of the cone of 95% confidence; dm and dp are the semimajor and semiminor axes, respectively, of the oval of 95% confidence.

* Rb-Sr whole-rock age is 109 ± 10 m.y. (Halpern, 1967).

† Rb-Sr whole-rock age is 103 ± 6 m.y. (Halpern, 1967).

** Average.

†† Calculated by method of Watson and Irving (1957).

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might indicate preferential removal of a secondary (possibly reversed) component of magnetization.

3. Thermal demagnetization to 550 °C of samples from four plutons, two dikes, and two localities in the porphyry fail to reveal the existence of any secondary directions of magnetization.

4. Microscopic observation of selected polished sections reveals no characteristics that indicate a common regional alteration. Magnetite crystals in both dikes and plutons display a complete range of alteration; some rocks have homogeneous euhedral grains of magnetite, while other rocks, apparently subjected to high-temperature hydrothermal alteration, have hematite and ilmenite lamellae parallel to (111) crystallographic planes in magnetite. About 25% of these samples show low-temperature production of maghemite. The magnetite crystals in the massive porphyry are almost universally altered to hematite, and they contain occasional, although far from ubiquitous, ilmenite lamellae. The ferromagnesian minerals in the rhyodacite porphyry and in most dikes display differing amounts of secondary chlorite and/or epidote formation, due either to late-stage deuteric effects during cooling or to hydrothermal alteration associated with plutonism.

5. The highly consistent radiometric ages on plutons and dikes in or near the sampled area are additional support of little post-emplacement thermal or chemical alteration.

The mean paleomagnetic direction derived from Upper Creta-

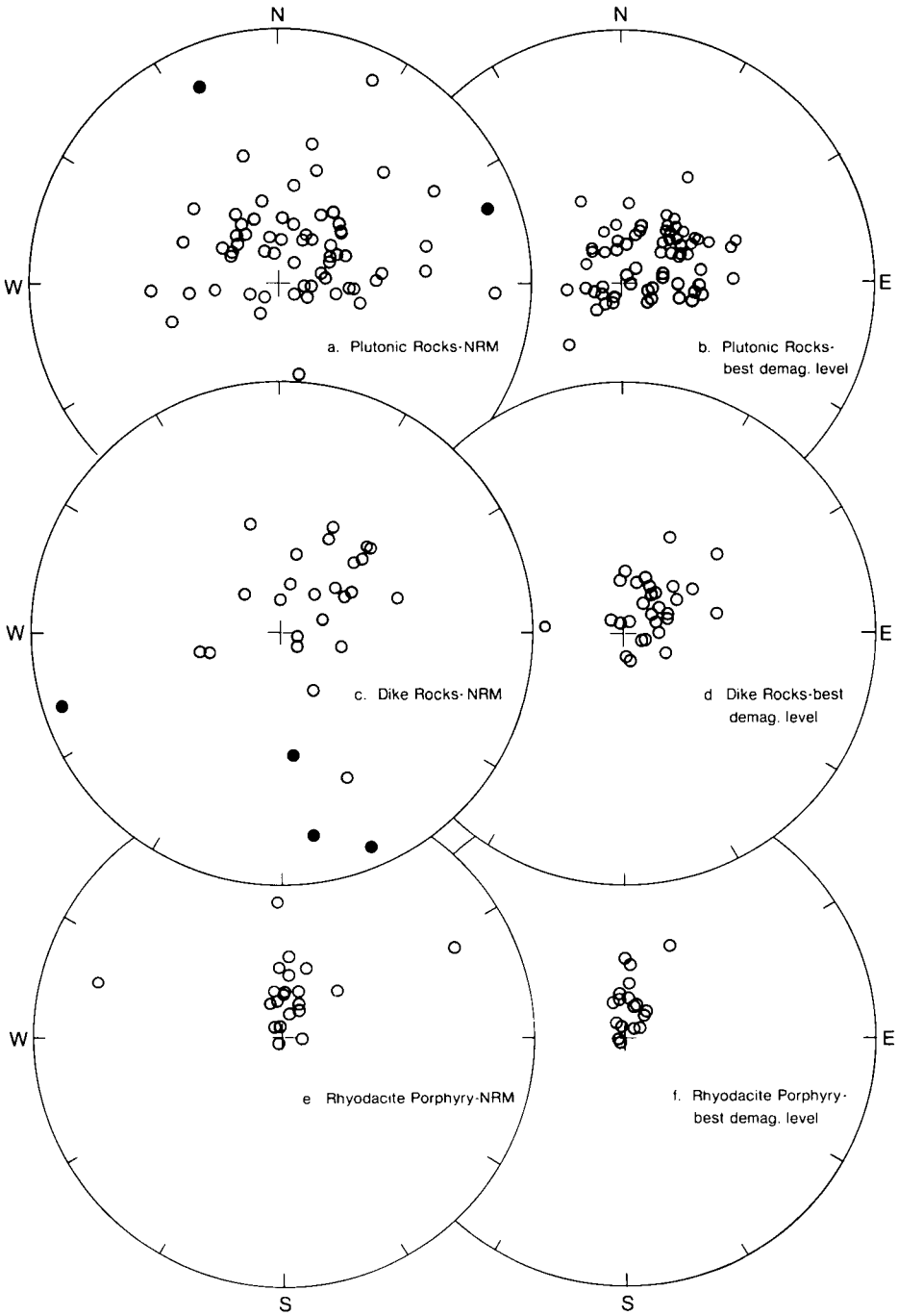


Figure 2. Equal-area projections showing natural remanent magnetization directions and directions at best demagnetization level for all samples from plutonic rocks (a, b), dikes (c, d), and rhyodacite porphyry (e, f). Open dots = upper-hemisphere projections; filled dots = lower-hemisphere projections.

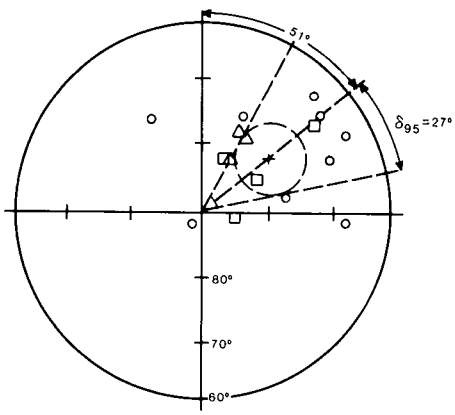


Figure 3. Central part of an equal-area projection (outer circle represents 60° inclination) showing site mean paleomagnetic collecting localities in Orville Coast and eastern Ellsworth Land. All directions are upper hemisphere. Circles = plutons, squares = dikes, and triangles = massive rhyodacite unit. Cross = the mean direction for all 92 samples from dikes and plutons, surrounded by circle of 95% confidence. This circle is tangent to limits in uncertainty in declination (δ_{95}).

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ceous rocks from the central and northern Lassiter Coast (Kellogg and Reynolds, 1978), is almost coincident with the present axial dipole field direction. In marked contrast to these Lassiter Coast data, nearly all mean directions from the Orville Coast and eastern Ellsworth Land show a declination to the northeast and are inclined steeply upward (Fig. 3). It should also be noted (Table 1) that there is no significant difference at the 95% confidence level between the mean directions derived from the nine stable plutons ($D = 49°$, $I = -77°$, $\alpha_{95} = 6°$), four dikes ($D = 56°$, $I = -77°$, $\alpha_{95} = 7°$) or the massive porphyry ($D = 30°$, $I = -79°$, $\alpha_{95} = 4°$). The significance of these uniform directions is discussed in the next section.

Because there is uncertainty about the structural and thermal histories of the area, as well as about the age of the porphyry, the mean Late Cretaceous (~100 m.y.) pole from this region is calculated for the plutons and dikes alone; it lies at 70.8°S, 164.4°W. The semimajor (*dm*) and semiminor (*dp*) axes of the oval of 95% confidence are 11.0° and 10.2°, respectively. If the four sites in the porphyry are included, however, the mean direction changes by less than 1°.

WHY SIMILAR DIRECTIONS FROM ALL ROCK TYPES?

The similarity in paleomagnetic direction between the plutons and dikes is readily understood, as they must be of approximately the same age. The close grouping of directions also suggests that cross-peninsular compression, and the resultant folding that characterizes the Latady Formation and the layered Jurassic volcanic rocks, had ceased prior to pluton and dike emplacement.

The tight grouping of all directions from the massive porphyry unit and the close similarity of these directions to those from other rocks is surprising, as the porphyry was originally suspected to be Late Jurassic in age and therefore more deformed than the plutons and dikes. There seem to be three possibilities to explain all the geological and paleomagnetic observations: (1) the massive porphyry has remained relatively undeformed since emplacement in Late Jurassic time, and the mean field direction changed little between Late Jurassic and Late Cretaceous time, (2) the porphyry is only slightly older than the intrusive rocks, thus also about Late Cretaceous in age, or (3) the directions of magnetization in the porphyry were magnetically reset by Late Cretaceous plutonism.

The first possibility appears unfeasible because of the unlikelihood that the mean apparent field direction remained unchanged between Late Jurassic and Late Cretaceous time. The second alternative also seems unlikely for several reasons. For example, there is a locally developed fracture cleavage in the porphyry parallel to well-developed axial-plane cleavage in the nearby folded sedimentary rocks, indicating that the porphyry was present during at least part of the period of folding. In addition, the porphyry is petrographically similar to highly folded flows that are interpreted from fossil evidence to be Late Jurassic in age (M.R.A. Thomson, 1978, oral commun.).

Although components of secondary magnetization could not be detected in the laboratory, thorough regional remagnetization seems to be the most feasible alternative to explain the similarities in direction between the porphyry and the intrusive rocks. The uniform normal polarity found in the porphyry and intrusive rock types lends support to this possibility; the pervasive alteration of the porphyry may well reflect this period of remagnetization.

The history of the porphyry might be better resolved if reliable paleomagnetic results could be obtained from some of the folded volcanic flows. Unfortunately, no paleomagnetically stable flows were collected, although a petrographically similar, steeply dipping Upper Jurassic flow from the Lassiter Coast yielded a stable paleomagnetic direction that was highly divergent from the results of either this study or that of the Andean rocks of the Lassiter Coast (Kellogg and Reynolds, 1978). If similar results were to be obtained for the flows of the Orville Coast and eastern Ellsworth Land, serious doubt would be placed on a history of remagnetization for the porphyry.

EVIDENCE FOR OROCLINAL BENDING OF THE SOUTHERN ANTARCTIC PENINSULA

The 51° mean northeast declination in paleomagnetic direction derived from the plutons and dikes leads to several conclusions about the regional tectonics. The uncertainty in this declination (δ_{95}) is ±27° (Fig. 3). Because paleomagnetic declinations from most other collecting localities in Andean rocks of the peninsula demonstrate no appreciable displacement from due north (Fig. 4), the results from the Orville Coast and eastern Ellsworth Land lead one to suspect major oroclinal bending for this region. The angular difference in the strike of tectonic fabric (defined by fold-axis directions, petrographic distributions, and shape of the peninsula) between the central and northern Lassiter Coasts and the study area varies between about 30° and 70° (K. S. Kellogg and P. D. Rowley, unpub. data; Kellogg and Rowley, 1974), in the same sense as the angular difference in paleomagnetic declination. This is strong support for the theory that the southernmost part of the peninsula has been bent relative to the rest of the peninsula since Late Cretaceous time.

The exact timing or geological history of this bending is not precisely known, although possible mechanisms have been hypothesized from detailed analyses of sea-floor magnetic patterns and basement structure in the southeastern Pacific. For example, Herron and Tucholke (1976) postulated that the Antarctic Peninsula,

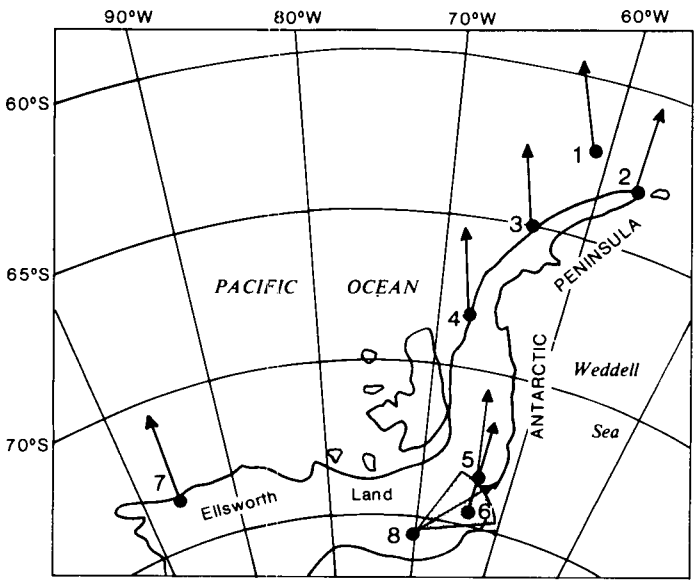


Figure 4. Declination of mean paleomagnetic vectors for West Antarctica Andean intrusive rocks, modified from Kellogg and Reynolds (1978). Sources of data: 1, Dalziel and others (1973); 2, 3, and 4, Blundell (1962); 5 and 6, Kellogg and Reynolds (1978); 7, Scharnberger (1980); 8, this study. Limits of uncertainty at 95% confidence level are indicated only for results from this study.

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including part of eastern Ellsworth Land, may have been a part of the East Antarctic plate since at least late Mesozoic time and that compression from the west occurred while the now-extinct Aluk plate (or possibly several smaller plates) was subducted beneath the western margin of the peninsula (Fig. 5). Western Ellsworth Land and Marie Byrd Land were then part of a separate West Antarctic plate, and a spreading center or ridge existed between the East and West Antarctic plates in what is now the Byrd Basin. With such a scenario, the base of the Antarctic Peninsula slid past the West Antarctic plate along a right-lateral transform fault. Frictional drag along this fault may in fact have produced the torque necessary to cause the observed rotation. These forces ceased when the Aluk Ridge, between the West Antarctic plate and the peninsula, subducted into its own trench, a process possibly still continuing at the northern end of the peninsula.

A similar scheme has been outlined by Molnar and others (1975), although they postulated that after initial Late Cretaceous (83 to 63 m.y. ago) spreading between East and West Antarctica, there followed a period of about 500 km of convergence that lasted until 38 m.y. ago.

Understandably, a few problems remain with such plate reconstructions. For one thing, any spreading between East and West Antarctica would have produced substantial volcanism if not oceanic crust in the Byrd Basin region. However, aeromagnetic surveys over this area (Behrendt, 1964) detect no significant volume of volcanic rocks, although a thick sedimentary cover could significantly mask any underlying volcanic rocks (J. C. Behrendt, 1978, oral commun.). Furthermore, the crustal thickness in the Byrd Basin is only slightly less than what would be suspected for normal continental crust (Bentley and others, 1960). LeMasurier (1980), on the basis of detailed petrologic investigations of post-Eocene volcanism in Marie Byrd Land, concluded that the deeply ice-buried region between East and West Antarctica represents the failed arm of a triple junction, a rift zone comparable to other continental rift zones of the world.

One might argue that the forces that caused the necessary oroclinal rotation might also produce second-order folding and possibly right-lateral slip faulting in the Orville Coast region. No such second-order folding has been detected, although many small displacement faults, predominantly apparent right-lateral strike slip, parallel to the postulated transform-fault orientation have been mapped in the region (K. S. Kellogg and P. D. Rowley, unpub. data).

It should be noted that the paleomagnetic data could indicate clockwise rotation of a relatively small fault-bounded crustal block rather than oroclinal bending of a large part of the peninsula. The large faults bounding such a block, if they exist, are apparently hidden by the extensive ice cover.

It is becoming apparent that the southern Antarctic Peninsula region is a crucial area of study for our understanding of plate-tectonic processes. Future detailed geological and paleomagnetic investigations, as well as geophysical probes into the nature of the crust under the Byrd Basin and Weddell Sea, should produce a fascinating array of information that will help to unfold the tectonic history of this complex area.

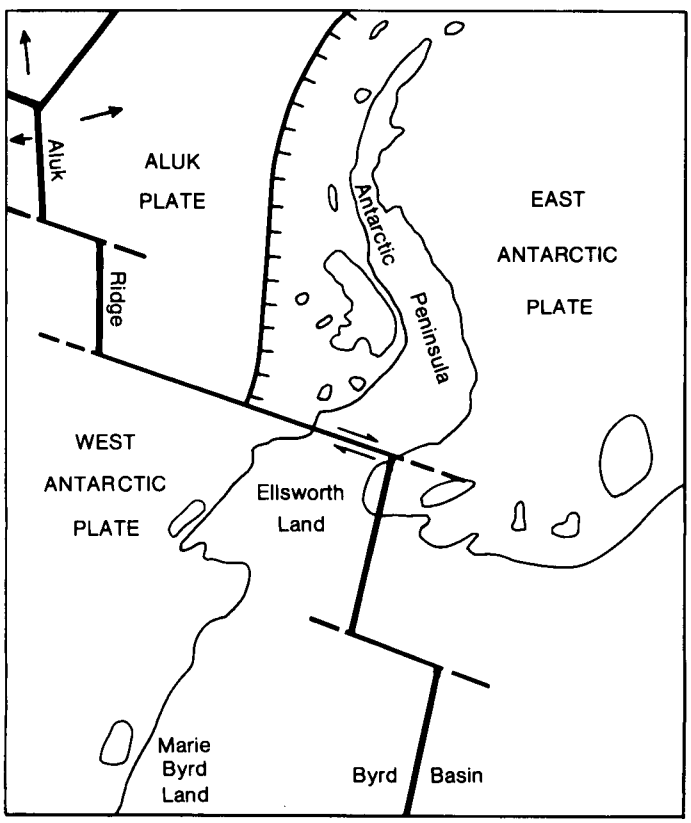


Figure 5. Possible plate configuration in Eocene to Miocene time, based on work of Herron and Tucholke (1976), using modern shoreline configuration. Trench (hachured line) off Antarctic Peninsula and inferred spreading center between East and West Antarctica (heavy line segments in lower part of figure) are now inactive. Arrows show direction of movement along postulated right-lateral transform fault.

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